Topic 3 Lab Question 1

$$\ddot{y} + 10\dot{y} + 21y = 0$$

$$\lambda^2 + 10\lambda + 21$$

$$(\lambda + 7)(\lambda + 3)$$

$$\lambda_2 = -7$$
$$\lambda_1 = -3$$

$$C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$$

$$C_1 e^{-3x} + C_2 e^{-7x}$$